MIL-F-36979A
3 July 1975
SUPERSEDING
MIL-F-36979
11 July 1973

MILITARY SPECIFICATION

FOOTWEAR COVER, DISPOSABLE, CONDUCTIVE, NONWOVEN FABRIC, UNIVERSAL SIZE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of a universal size, nonwoven fabric, conductive, disposable footwear cover.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-276	Thread, Cotton
PPP-B-636	Boxes, Shipping, Fiberboard

STANDARDS

Federal

FED-TEST-METHOD STD NO. 191	Textile Test Methods
FED-STD-751	Stitches, Seams, and Stitchings

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 8430

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE CODES

(Application for copies should be addressed to the National Fire Protection Association (NFPA), 470 Atlantic Avenue, Boston, Massachusetts 02210.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials and components. The materials and components shall conform to the applicable specifications and standards as listed or required herein.

3.1.1 Footwear cover fabric. The footwear cover shall be fabricated from nonwoven, moisture-proof, spun bonded, olefin fabric. The color of the face of the fabric or the outerside of the footwear cover shall be green, blue green, or blue either in striped effect or solid color. The underside of the fabric or the innerside of the cover shall be white or the natural color of the olefin material. When inspected as specified in 4.3.1 and 4.3.1.1, the footwear cover fabric shall conform to the requirements of Table I.

TABLE I
FOOTWEAR COVER FABRIC PROPERTIES

CHARACTERISTICS	REQUIREMENTS
Ounces per square yard, minimum	1.2
Breaking strength, pounds minimum:	
Machine direction	6
Cross direction	9

3.1.2 Conductive stripping. The conductive stripping shall be fabricated from woven or nonwoven fabric durably impregnated with conductive carbon. When inspected as specified in 4.3.2 through 4.3.2.2, the stripping shall conform to the requirements of Table II.

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TABLE II
CONDUCTIVE STRIPPING PROPERTIES

CHARACTERISTICS	REQUIREMENTS
Ounces per yard, minimum	1.6
Breaking strength-full width, pounds minimum	13
Electrical resistance, ohms maximum	500,000
Width, inches minimum	11/16
Color	Black

3.1.3 Elastic binding. The elastic binding shall be fabricated from natural or synthetic rubber covered with 2 or 3 ply cotton yarn or shall be a strip of natural or synthetic rubber. The color of the binding shall be natural or a neutral shade. The binding shall be of a suitable width and of sufficient elasticity and strength suitable for use as a component of the footwear cover. A certificate of quality may be submitted (see 4.1.2.).

3.1.4 Thread. All the sewing operations shall be performed with the cotton thread conforming to V-T-276, type 1A1, or 1A2, Ticket Number 40, 50 or 60, 3 ply. As an alternate Number 7, 4 ply stretch nylon thread with 34 filaments per ply and Number 15, 2 ply stretch nylon thread with 34 filaments per ply may be substituted for the cotton thread. The stretch nylon thread shall be suitable for sewing the footwear cover. A certificate of quality for the stretch nylon thread may be submitted (see 4.1.2).

3.2 Design. The footwear cover shall be a one piece construction, closed back style; the conductive stripping on the bottom of the cover, with or without the elastic binding and with a heel opening for inserting the conductive stripping on the inside of the cover or without a heel opening and extending the stripping along the back of the cover, for inserting into the wearer's stockings; and elastic binding around the cover opening to hold the cover firmly on the wearer's shoes and ankles. The footwear cover shall conform in appearance to Figure 1 or 2, as applicable.

3.3 Construction.

3.3.1 Seams and stitching. The seam and stitch types used in the construction of the footwear covers shall conform to FED-STD-751. The overedge stitching shall be stitch type 503, 504 or 505. The bight of the overedge stitching shall be 1/8 to 1/4 inch from the edge of the fabric. All regular sewing shall be performed with Stitch Type 301. All sewing shall be accomplished with a minimum of 5 stitches per inch. The seam allowance may be on the inner or outerside of the cover. Each

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thread break, skip, and run-off shall be overstitched not less than 1/2 inch. Skipped stitches or thread breaks on Stitch Type 500 Class stitching may be repaired using Stitch Type 301. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together. All thread ends shall be trimmed to a maximum length of 1/4 inch.

3.3.2 Construction details. The top of the cover from the front of the opening, along the instep-toe area and on the bottom or sole area to the heel area shall be closed with Seam Type SSa-1 and overedge Stitch Type 500 class. The elastic binding and conductive stripping shall be attached to the bottom of the cover, starting 1-1/2 ± 1 inches from the tip of the top when the cover is in a natural flattened state as it would be when worn. The elastic binding may be attached to the inner or outerside of the cover. The closing of the sole and the attachment of the elastic binding and conductive stripping may be accomplished in one operation. The conductive stripping shall extend beyond the heel attachment point 12 ± 1 inches to permit insertion inside the wearer's hosiery or between the shoe and hosiery to assure conduction or grounding of static electricity. The conductive stripping, when on the outerside of the cover at the heel portion, may or may not be attached (sewn) to the back of the heel. The cover may have an opening, 3/4 ± 1/4 inch, at the heel area for inserting the conductive stripping to the inside of the cover. The heel opening may be omitted at the option of the manufacturer. As an alternate, the cover may be fabricated without any seam or elastic binding on the bottom of the cover. The conductive stripping may be attached flat on the bottom cover and extend onto the toe area for approximately one inch. The stitching attaching the flat conductive stripping shall be approximately in the center of the stripping, or approximately 1/4 inch from one edge. The elastic binding shall be attached around the circumference of the cover opening with Stitch Type 500 class. The assembled cover when stretched to its extreme, without breaking the stitching or fabric, shall be a minimum of 14 inches in the overall length, a minimum of 4-1/2 inches in the heel height and a minimum diameter of 7-1/2 inches for the cover opening.

3.4 Performance inspection.

3.4.1 Electrical resistance. The electrical resistance of each assembled cover shall be not greater than 500,000 ohms when inspected as specified in 4.3.4.2.

3.5 Workmanship. The covers shall not contain any metal, spot, stain, non-specified hole, abraded area, tear, cut, mend, patch, scorch or needle chew. The covers shall conform to the quality of grade established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material and component inspection. In accordance with 4.1, the materials and components shall be inspected and tested in accordance with all the requirements of the referenced specifications and standards, unless otherwise excluded, amended, modified, or qualified in this specification or in the purchase document.

4.1.2 Certificate of quality. When certificates of quality are submitted in accordance with Table III, they shall contain verifiable actual test and inspection data. The Government reserves the right to inspect and test the components to verify the validity of the certification.

4.1.3 Inspection. Inspection, as used in this specification, is defined as both examination (an investigation without the use of special laboratory apparatus or procedures such as visual) and testing (a determination by technical means of the physical and chemical properties) of the item.

4.2 Quality conformance inspection. The sampling and inspection shall conform to MIL-STD-105, except where otherwise indicated hereinafter. The quality conformance inspection shall consist of the tests and examinations specified in Table III.

4.2.1 Sampling.

4.2.1.1 Inspection lot.

4.2.1.1.1 Cover fabric. An inspection lot size shall be expressed in units of one linear yard of the cover fabric and shall consist of all the cover fabric received by the cover manufacturer at one time. The sample unit for inspection shall be one linear yard.

4.2.1.1.2 Conductive stripping. An inspection lot size shall be expressed in units of one linear yard of the conductive stripping and shall consist of all the conductive stripping received by the cover manufacturer at one time. The sample unit for inspection shall be one linear yard.

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4.2.1.1.3 Elastic binding. An inspection lot size shall be expressed in units of one linear yard of the elastic binding and shall consist of all the elastic binding received by the cover manufacturer at one time. The sample unit for inspection shall be one linear yard.

4.2.1.1.4 Footwear covers. An inspection lot size shall be expressed in units of one footwear cover made essentially under the same conditions and from the same materials and components. The sample unit for inspection shall be one footwear cover.

4.2.1.1.5 Preparation of delivery. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing footwear covers, fully prepared for delivery from essentially the same materials and components. The sample unit for inspection shall be one shipping container, containing footwear covers, fully prepared for delivery with the exception that it need not be sealed.

4.2.1.2 Sampling for tests and examinations of the cover fabric, conductive stripping, elastic binding, footwear covers and preparation for delivery. The sample size, acceptance criteria, tests and examinations required for the cover fabric, conductive stripping, elastic binding, footwear covers, or preparation for delivery, as applicable, shall be as specified in Table III.

TABLE III

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATIONS REQUIRED FOR THE COVER FABRIC, CONDUCTIVE STRIPPING, ELASTIC BINDING, FOOTWEAR COVERS AND PREPARATION FOR DELIVERY

ITEM	INSPECTION	PARAGRAPH REQUIREMENT	PARAGRAPH METHOD	SAMPLE SIZE	ACCEPTANCE CRITERIA	
Cover Fabric 1/	Weight	3.1.1	4.3.1	2/	2/	3/
	Breaking strength	3.1.1	4.3.1	2/	2/	3/
	Material	3.1.1	4.3.1.1	2/	2/	3/
	Color	3.1.1	4.3.1.1	2/	2/	3/
Conductive Stripping 1/	Weight	3.1.2	4.3.2	2/	2/	3/
	Breaking strength	3.1.2	4.3.2	2/	2/	3/
	Electrical Resistance	3.1.2	4.3.2.1	2/	2/	3/
	Width	3.1.2	4.3.2.2	2/	2/	3/
	Color	3.1.2	4.3.2.2	2/	2/	3/

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TABLE III (Cont'd)
SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATIONS REQUIRED FOR THE COVER FABRIC, CONDUCTIVE STRIPPING, ELASTIC BINDING, FOOTWEAR COVERS AND PREPARATION FOR DELIVERY

ITEM	INSPECTION	PARAGRAPH REQUIREMENT	PARAGRAPH METHOD	SAMPLE SIZE	ACCEPTANCE CRITERIA
Elastic Binding 1/	Kind of Rubber	3.1.3	4.3.3	2/	2/ 3/
	Material of covering yarn	3.1.3	4.3.3	2/	2/ 3/
	Ply of covering yarn	3.1.3	4.3.3	2/	2/ 3/
	Color	3.1.3	4.3.3	2/	2/ 3/
Footwear Covers	Visual Examination	3.2 through 3.3.2 and 3.5	4.3.4.1.1	Inspection Level S-2	An acceptable quality level of 4.0 defects per 100 units for major defects and 10.0 defects per 100 units for minor defects.
	Electrical Resistance	3.4.1	4.3.4.2	Inspection Level S-1	An acceptable Quality level of 1.5 defects per 100 units
	Preparation for Delivery	Section 5	4.3.4.1.2	Inspection Level S-2	An acceptable quality level of 2.5 defects per 100 units

1/ The inspection of the cover fabric, conductive stripping, and the elastic binding, as applicable, shall be conducted either at the contractor's or sub-contractor's plant.

2/ The sampling for the testing of the cover fabric, conductive stripping and the elastic binding shall be in accordance with 4.2.1.1.1 through 4.2.1.1.3, as applicable, and as follows:

Lot Size (Yards)	Sample Size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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The samples shall be taken from the beginning of each roll. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified in Table III. Each sample unit taken for inspection shall be used to determine all the characteristics of the cover fabric, conductive stripping and the elastic binding, as applicable, specified in Table III.

3/ A certificate of quality shall be submitted and will be acceptable for the stated requirement (see 4.1.2).

4.3 Inspection methods.

4.3.1 Cover fabric. The characteristics of the cover fabric shall be determined in accordance with the inspection methods of FED-TEST METHOD-STD. NO. 191 specified in Table IV and 4.3.1.1.

TABLE IV
COVER FABRIC PROPERTIES

Characteristic	FED-STD-191 Method
Weight	5041
Breaking strength	5102

4.3.1.1 Material and color. The composition of the material of the cover fabric shall be determined by any standard commercial standard. The color of the fabric shall be determined visually.

4.3.2 Conductive stripping. The conductive stripping characteristics shall be determined in accordance with the inspection methods of FED-TEST METHOD-STD NO. 191 specified in Table V, 4.3.2.1 and 4.3.2.2.

TABLE V
CONDUCTIVE STRIPPING PROPERTIES

Characteristic	FED-STD-191 Method
Weight	5041
Breaking strength (full width)	5102

4.3.2.1 Electrical resistance. The electrical resistance properties of the conductive stripping shall be determined in accordance with NFPA-56A.

4.3.2.2 Width and color. The width of the conductive stripping shall be determined with the use of any suitable measuring device. The color of the stripping shall be determined visually.

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4.3.3 Elastic binding. The kind of rubber and ply of the yarn, when applicable, of the elastic binding shall be determined by any standard commercial method. The material of the covering yarn, when applicable, of the elastic binding shall be determined in accordance with FED-TEST METHOD-STD NO. 191, Method 1200. The color of the yarn shall be determined visually.

4.3.4 Footwear covers.

4.3.4.1 Visual examination.

4.3.4.1.1 Footwear covers. Each footwear cover selected in accordance with Table III as a sample unit from the lot shall be thoroughly checked dimensionally and examined visually to determine conformance to this specification. The classification of defects, Tables VI, shall be used to classify the defects found.

TABLE VI
CLASSIFICATION OF DEFECTS FOR THE VISUAL EXAMINATION OF THE ASSEMBLED FOOTWEAR COVERS

	DEFECT	MAJOR	MINOR
a.	Any hole, scissors or knife cut, tear, mend, split, burn, needle chew, abraded area, or any other weakening defect	X	
b.	Any portion of the cover stiffened or scorched by any process of manufacturing	X	
c.	Any cover containing any metal	X	
d.	Any part shaded or any shade bar		X
e.	Color of any component not as specified or not uniform		X
f.	Two or more thread ends not trimmed to a maximum length of 1/4 inch or three or more thread scraps not removed		X
g.	Any spot or stain on the outerside of any cover		X
h.	Any dimension not as specified	1/	
i.	Any thread break, skip or run-off not overstitched	X	
j.	Any thread break, skip or run-off overstitched for less than 1/2 inch	1/	

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TABLE VI (Cont'd)
CLASSIFICATION OF DEFECTS FOR THE VISUAL EXAMINATION
OF THE ASSEMBLED FOOTWEAR COVERS

DEFECT	MAJOR	MINOR
k. Any seam or stitch type not as specified	1/	
l. Any loose stitching 2/	1/	
m. Any tight stitching, resulting in the breakage of the stitches when normal pull is applied 3/	1/	
n. One or more stitches per inch less than the minimum specified	X	
o. Any component not as specified or any defect of component or defect of assembly, not herein classified	1/	
p. Any component, component part, or required operation omitted, or any operation improperly performed, not herein classified	1/	
q. Any open seam 4/	X	

1/ The defect shall be classified and scored as a major defect, when it seriously affects the serviceability or appearance, otherwise it shall be classified and scored as a minor defect.

2/ A seam shall be classified and scored as loose when the top and bobbin threads are on the same surface.

3/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident and is not caused by the gathering of the material, the stitching shall be inspected for tight tension and breaking of stitches by exerting normal pull in the lengthwise direction of the stitching, by pulling the material taut to straighten out the seam.

4/ A seam shall be classified and scored as open when one or more stitches joining a seam are broken or when two or more continuous skipped stitches or run-offs occur.

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4.3.4.1.2 Preparation for delivery. Each of the fully prepared shipping containers, containing footwear covers, selected in accordance with Table III as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects, Table VII, shall be used to enumerate the defects found.

TABLE VII
LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

ITEM	DEFECT
Containers	Not as specified.
Quantity	Quantity in shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size location, sequence, or method of application

4.3.4.2 Electrical resistance. The electrical resistance properties of the footwear covers shall be determined in accordance with NFPA-56A.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Unit of issue. One box containing 200 foot covers, as specified, constitutes one unit of issue.

5.1.2 Level A. Two hundred foot covers shall be neatly folded and packed in a snug fitting box constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of PPP-B-636.

5.1.3 Level B. Two hundred foot covers shall be packed in a snug fitting box constructed in accordance with PPP-B-636, class domestic. Closure shall conform to method I or II as specified in the appendix of PPP-B-636.

5.1.4 Level C. Two hundred covers shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.2 Marking.

5.2.1 Exterior container. Each exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The conductive nonwoven fabric footwear covers are intended for one time short term use by nonregular vistors to surgical suites or other areas where flammable.liquids and gases are used.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Quantity desired.
c. Selection of applicable levels of packing (see 5.1).

6.3 Catalog information. This specification is intended to cover the following item appearing in the Federal Supply Catalog:

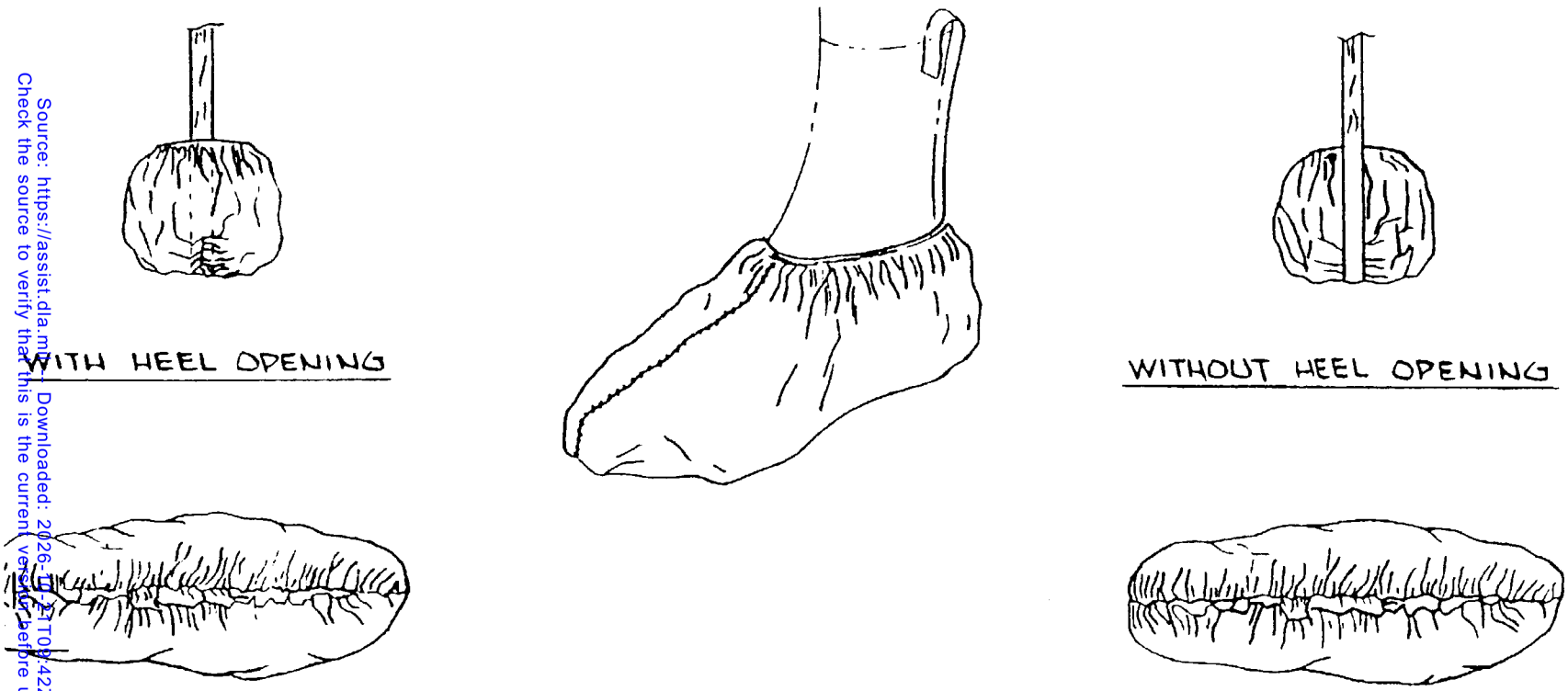
National Stock No.	Item Identification
8430-00-186-3450	FOOTWEAR COVER, DISPOSABLE, Conductive, Nonwoven Fabric, Universal Size

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 8430-0223
CS

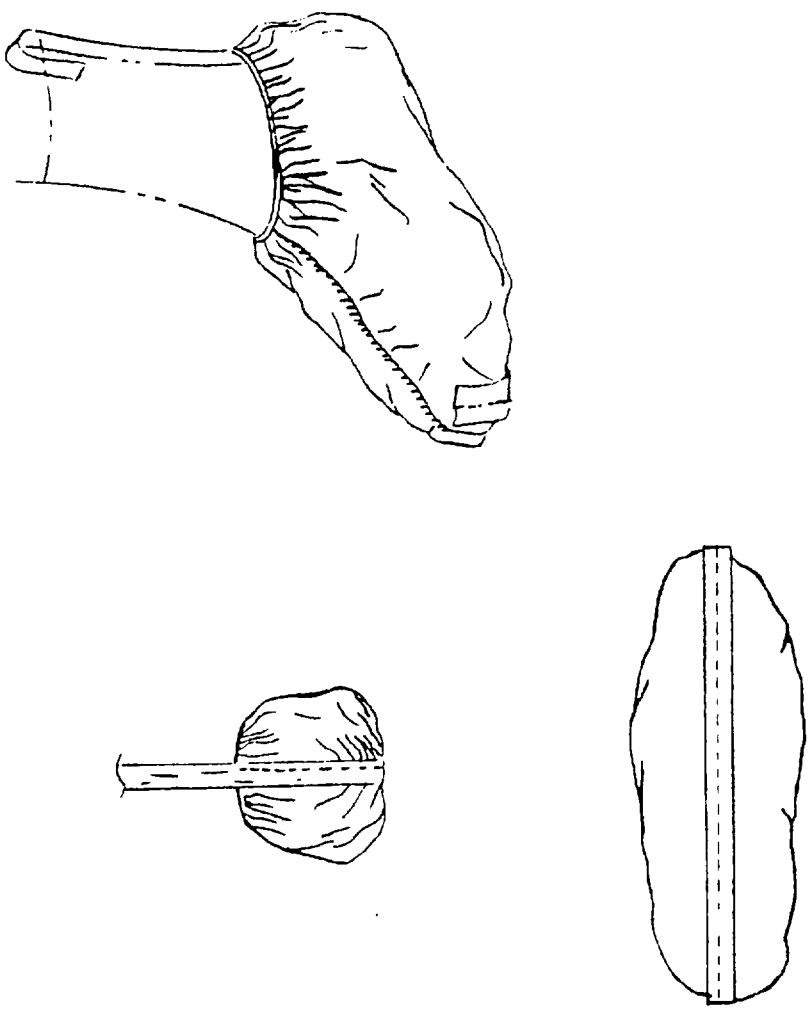
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FOOTWEAR COVER, DISPOSABLE, CONDUCTIVE, NONWOVEN FABRIC

FIGURE 1

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FOOTWEAR COVER, DISPOSABLE, CONDUCTIVE, NONWOVEN FABRIC

FIGURE 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-F-36979, FOOTWEAR COVER, DISPOSABLE, Conductive, Nonwoven Fabric, Universal Size

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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